



NATIONAL TECHNICAL UNIVERSITY OF ATHENS
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING KNOWLEDGE AND DATA
SYSTEMS LABORATORY

Ak. year 2023-2024, 6th semester, ECE

Databases
Semester Work

I. Pronunciation

Popular cooking competition asked you to design and implement a system for storing and managing the information required for the operation of the competition regarding the recipes, ingredients and equipment required for the operation of the competition.

Through the relevant application, he wishes to handle cooking and pastry recipes. Each recipe is a cooking or pastry recipe. It also belongs to a national cuisine¹ (eg French, Italian) and has a level of difficulty (eg very easy = 1 easy = 2, medium = 3, difficult = 4, very difficult = 5).

Each recipe has a name, a short description and belongs to one or more forms of meal (eg breakfast, breakfast, lunch, afternoon, dinner). The competition would also like each recipe to be categorized through additional, unlimited, labels for the form / type of meal (eg brunch, quick-lunch, cold dish, etc.) and possibly have up to 3 useful tips (eg kept in the refrigerator for up to 3 days).

The execution of a recipe requires specific equipment / accessories (eg bowl, sieve, blender, whisk) regardless of quantities (eg 3 blenders). Each accessory / equipment has instructions for use (eg we use the wire when we want to give a mixture volume).

The total time required to execute a recipe is broken down into preparation time and cooking time. Each recipe consists of 1 or more steps, which must be performed sequentially. Each step describes what the cook should do (e.g. in a bowl, put the eggs and whisk them). After all the steps have been performed, specific quantities will arise (eg 2 servings).

Each recipe requires specific ingredients in quantities, clearly or less clearly defined (eg 100 g feta cheese, 2 eggs, 1 tsp oil, a little pepper, a little flour). The ingredients used in recipes (foods) are grouped into Food Groups. For food groups you can use either the National Food and Drink Code² - or the Institute of Preventive, Environmental and Occupational Medicine³. Each food group has a name and description.

Each recipe has a single ingredient that is classified as basic. Each recipe is characterized by its basic ingredient (e.g. whether the basic ingredient belongs to the food group Various foods

¹ https://el.wikipedia.org/wiki/Category:National_cuisines

² 12 categories (e.g. Fats and oils, Sweeteners, Meat) <https://el.wikipedia.org/wiki/Food>

³ 10 κατηγορίες (πχ λαχανικά, όσπρια) <http://www.diatrofikoiodigoi.gr/?Page=omades-trofimon-sub>

then the recipe is characterized as vegetarian, if it is the basic ingredient in the category "Fish and fish products" then the recipe is characterized as seafood).

Each recipe can have nutritional information. These are grams of fat per serving, grams of protein per serving, grams of carbohydrates per serving, and number of calories per serving. The calories of each serving should be calculated dynamically based on the individual calories and proportions of ingredients, as the ingredients / foods have a specific number of calories⁴ per 100 g or ml (eg strawberries 24 calories / 100 g).

The competition groups, according to its own criteria, the recipes into thematic sections (eg village recipes, risotto recipes, Easter sweets). Each recipe can be included in one or more of such sections. Each section has a name (eg Easter sweets) and description (eg Easter sweets ideal for the Easter table).

Each recipe has one or more cooks⁵ who can execute a recipe. Each cook has a name, surname, contact number, date of birth, age, years of professional experience and specialization in one or more national cuisines. It also has a vocational training qualification as follows: C' cook, B' cook, A' cook, assistant chef, chef (chef).

The competition takes place annually in 10 episodes/reruns. In each episode, 10 national cuisines, 10 representative cooks from each kitchen, 3 judges and 1 recipe from each national cuisine assigned to 1 cook are automatically selected by the system. (A cook/judge/national cuisine/recipe cannot participate continuously in more than 3 episodes; also each cook participating in the competition should be assigned the recipe he is asked to perform.)

Each cook executes the recipe assigned to him and is graded by the 3 judges, based on an integer numerical scale of 1 5. The cook who will get the highest score is winner. In the event of a tie, the winner is the cook with the highest professional training. In the event of a new tie, the winner is declared random.

All entities stored in your app should be accompanied by corresponding images (e.g. images for food groups, sections/ recipes/ ingredients/ cooks/ episodes, images for equipment/ components, etc.). Each image should have a verbal description of what it depicts (eg Banana protein smoothie).

- Application Users: For each user, the system must verify his/her identity when accessing the application (via username / password).
 - Administrator: Registers and modifies all required information. It can back up the entire database (backup) and restore the system from it (restore).
 - Cook: Cooks have the ability to edit all the elements of the recipes assigned to them and also add a new recipe. They can also

⁴ <https://diatologos.com/thermidometritis/>

⁵ <https://el.wikipedia.org/wiki/Μάγειρας>

process their personal information. They cannot modify other elements of the system, e.g. recipes that have not been assigned to them.

The specifications of the data/information as well as the reports may not have been sufficiently defined by the competition. For the sake of completeness of your work you should record in detail the specifications of the system as well as your assumptions.

You will need to enter information about each of the entities in the DB. In the DB there should be enough data to successfully execute all requested queries and return the appropriate information. *If running a query does not return data, then the corresponding query will not be scored.* Indicatively, there should be more than 50 recipes, 100 ingredients, 50 cooks, 50 episodes. You can get information about recipes from the <https://www.airtable.com/universe/expHZcS7kWEyq5gUH/recipe-database>

1. (10%) Draw the ER diagram resulting from the above description.
2. (20%) Draw the relational diagram and develop the DB.
 - 2.1. (5%) Define all necessary constraints to ensure the correctness of the DB. These are integrity limitations, keys, referential integrity, value field integrity, and user-defined limitations.
 - 2.2. (5%) Define appropriate indexes for the tables in the database and justify your choice based on their usefulness for the queries in which they are used.
3. Through the application a user will be able to execute and see the results for the following queries: (60% - queries are equal)
 - 3.1. Average Ratings (score) per cook and National cuisine.
 - 3.2. For a given National cuisine and year, which cooks belong to it and which cooks participated in episodes?
 - 3.3. Find the new cooks (age < 30) who have the most recipes.
 - 3.4. Find cooks who have never participated as judges in an episode.
 - 3.5. Which judges have participated in the same number of episodes over a year with more than 3 appearances?
 - 3.6. Many recipes cover more than one label. Among the pairs of fields (e.g. brunch and cold dish) common to recipes, find the top 3 pairs that appeared in episodes⁶ For this question, your answer should include, in addition to the query, an alternative Query Plan (eg with force index), the corresponding traces and your conclusions from their study.
 - 3.7. Find all cooks who participated at least 5 fewer times than the cook with the most episode entries.
 - 3.8. In which episode were the most components (equipment) used? Similarly with question 3.6, your answer should include, in addition to the query, an alternative Query Plan (eg with force index), the corresponding traces and your conclusions from their study.
 - 3.9. List with average number of grams of carbohydrates in the competition per year?

⁶<https://mariadb.com/kb/en/index-hints-how-to-force-query-plans/>, <https://mariadb.com/kb/en/optimizer-trace-guide/>

- 3.10. Which National cuisines have the same number of entries in competitions, over two consecutive years, with at least 3 entries per year?
- 3.11. Find the top-5 judges who have given the highest overall score to a cook. (judge's name, cook's name and overall scoring score)
- 3.12. What was the most technically difficult, in terms of recipes, episode of the competition per year?
- 3.13. Which episode received the lowest degree of professional training (judges and cooks)?
- 3.14. Which theme has appeared the most times in the competition?
- 3.15. What food teams have never appeared in the competition?

II. Procedural

- Free choice of groups up to a **MAXIMUM of 3 people** (individual selection is allowed) **no later than 1/4/2024. Late registrations and/or modifications will not be accepted.**
- For the development of your application it is recommended to use MySQL (MariaDB) or Postgres to implement the DB. You can use PHP or Java or Python or Node JS to implement application points (eg competition draw).
- Work developed using ORM technologies will not be accepted.
- Queries must be implemented elegantly and efficiently in SQL.
- Copying is strictly prohibited. In case of copying (even partial), all copies of the same work will be graded zero.
- In addition to the above requirements, you are free to make your assumptions and justify them in order to reach the realization of your work.

III. Deliverable

Job report (until 26/5/2024). **Late reports and/or amendments will not be accepted.** The report should contain:

1. ER & Relational diagram of the NW with justification and indexes
2. DDL and DML script (without them step 2 cannot be evaluated 30%)
3. Detailed steps to install your app and any libraries required. All the configuration files needed to install your application from scratch.
4. Link to your app's git repo. If it is non-public, you must have granted us access by the date of delivery. In addition to the above, the source code of your application should also be located in the repo.
5. All submissions are strictly subject to the code of academic ethics of NTUA and SECE. Your code must not change from the day of delivery of the report until the grading of the course. If this happens, your score will be ZERO (0). The assignment is given exclusively within the framework of the teaching semester and can be delivered during the examination period of June only. Also, there is no review/rescoring of assignments within the same academic year.

IV. Demo App

The examination of the application will take place **before the exam** (an announcement will be made for the exact date). You should have the application installed on your computer (better to come with your own laptop) as well as a printed report (**at least printed Relational diagram and**

DDL script). No grade will be assigned to a group or student who does not attend the demonstration of the project.

V. Time limits

All dates are due and no extension will be given.

N/A	Date	Event
1.	1/4/2024	Group Statement
2.	26/5/2024	Deliverable
3.	27 - 31/5/2024	Demo App